Holographic Bulk Spacetime Reconstruction in Vaidya-AdS:

A Comparative Study of Neural Network Parametrizations vs. Orthogonal Spectral Bases and Sensitivity Analysis

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Abstract

We present a rigorous comparative study of two computational paradigms for solving the inverse holographic problem in $(2+1)$-dimensional Vaidya-AdS spacetime: reconstructing the dynamic bulk mass profile $m(v)$ from boundary entanglement entropy observables (regularized spacelike geodesic lengths L_{reg} and boundary interval widths h). The baseline implementation parameterizes the metric using an unconstrained multilayer perceptron (MLP) trained via reverse-mode automatic differentiation (discrete adjoint) through an unrolled Runge-Kutta integrator. We demonstrate that this neural network formulation suffers from critical pathologies: severe late-time asymptotic drift ($m(5) \approx 1.77$ vs. true 1.00), ill-conditioned non-convex loss landscapes ($\kappa \sim 10^8 - 10^{14}$), and the infeasibility of forward sensitivity analysis ($\sim$ 7,000 coupled ODEs). To resolve these deficiencies, we propose a compact spectral framework expanding $m(v)$ in an orthogonal Chebyshev polynomial basis over a compactified time coordinate $\xi = \tanh(v/v_0)$ modulated by a $\text{sech}^2(v/v_0)$ shell envelope. The spectral model preserves the exact vacuum-to-BTZ asymptotic limits ($m(-\infty) \equiv 0, m(+\infty) \equiv 1$) by construction, renders the metric strictly affine in the parameters, and enables millisecond forward sensitivity analysis via $J = \partial L/\partial \mathbf{c}$. Coupling this Jacobian to a Damped Gauss-Newton solver yields quadratic convergence ($10^{-3} \to 10^{-10}$ MSE) in 4 to 8 iterations (< 1 second) without pre-training. Finally, we benchmark both methods across smooth, multi-step, and pathological collapse regimes (including shockwave discontinuities, null energy condition violations, and turbulent high-frequency ripples), identifying fundamental holographic screening and Gibbs phenomena.

Keywords: AdS/CFT correspondence, Holographic entanglement entropy, Vaidya-AdS, Differentiable physics, Sensitivity analysis, Chebyshev spectral methods, Gauss-Newton optimization.

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1 Introduction and Theoretical Framework

In gauge/gravity duality, the Ryu-Takayanagi (RT) proposal [1] and its covariant Hubeny-Rangamani-Takayanagi (HRT) generalization [2] assert that the entanglement entropy S_A of a spatial boundary subregion A in a conformal field theory (CFT) is dual to the proper volume/area of a bulk extremal codimension-2 spacelike surface γ_A homologous to A:

$$S_A = \frac{\text{Area}(\gamma_A)}{4G_N}. \tag{1}$$

In $(2+1)$-dimensional spacetime, the extremal surface is a one-dimensional spacelike geodesic curve γ_A.

We consider gravitational collapse modeled by the $(2+1)$-dimensional ingoing Vaidya-AdS metric in Eddington-Finkelstein coordinates (v, r, x):

$$ds^2 = -f(r,v)\,dv^2 + 2\,dv\,dr + r^2\,dx^2, \qquad f(r,v) = r^2 - m(v), \tag{2}$$

where v is the ingoing advanced null coordinate, $r \in (0, \infty)$ is the holographic radial coordinate ($r \to \infty$ corresponds to the CFT boundary), and $x \in \mathbb{R}$ is the planar spatial coordinate. The function $m(v)$ denotes the dynamic mass profile. In the early-time limit ($v \to -\infty$), $m(v) \to 0$ corresponds to pure vacuum AdS_3, while in the late-time limit ($v \to +\infty$), $m(v) \to m_f = 1$ describes the static BTZ black hole of horizon radius $r_h = \sqrt{m_f} = 1$ [3].

1.1 Spacelike Geodesic Equations

A spacelike geodesic parameterized by an affine parameter $\lambda \in [-\lambda_{\max}, \lambda_{\max}]$ satisfies the Euler-Lagrange equations derived from the Lagrangian $\mathcal{L} = \frac{1}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$:

$$\ddot{v} = r\dot{x}^2 - \frac{1}{2}\frac{\partial f}{\partial r}\dot{v}^2, \tag{3}$$

$$\ddot{r} = f\ddot{v} + \frac{\partial f}{\partial r}\dot{r}\dot{v} + \frac{1}{2}\frac{\partial f}{\partial v}\dot{v}^2, \tag{4}$$

$$\ddot{x} = -\frac{2}{r}\dot{r}\dot{x}, \tag{5}$$

where dots denote derivatives with respect to affine parameter λ. Due to reflection symmetry $x \to -x$, the geodesic has a minimal turning point at $\lambda = 0$:

$$v(0) = v_0, \quad r(0) = r_\star, \quad x(0) = 0, \quad \dot{v}(0) = 0, \quad \dot{r}(0) = 0, \quad \dot{x}(0) = \frac{1}{r_\star}, \tag{6}$$

satisfying unit spacelike normalization $\dot{s}^2 = -f\dot{v}^2 + 2\dot{v}\dot{r} + r^2\dot{x}^2 = 1$ at $\lambda = 0$.

1.2 Holographic Boundary Observables

Integrating the half-geodesic outward to a UV radial cutoff $r_{\text{cut}} \gg 1$ yields two key boundary observables:

1. **Boundary spatial interval width:** $h = 2x(r_{\text{cut}})$.
2. **Regularized proper length:**

$$L_{\text{reg}} = 2\int_0^{\lambda_{\text{cut}}} \sqrt{-f\dot{v}^2 + 2\dot{v}\dot{r} + r^2\dot{x}^2}\, d\lambda - 2\ln(2r_{\text{cut}}). \tag{7}$$

1.3 The Inverse Problem

Given a target entanglement entropy dataset $\mathcal{D} = \{(h_i, L_{\text{reg},i})\}_{i=1}^M$ measured across various boundary intervals and observation times, the goal of holographic reconstruction is to invert the nonlinear functional map:

$$(h, L_{\text{reg}}) \xmapsto{\text{Inverse HRT}} m(v). \tag{8}$$

2 The Neural Network Approach

The original implementation in the repository (`HoloML/Vaidya_AdS/scripts/fit_metric_from_lreg.py`) represents the dynamic mass profile using a multilayer perceptron (MLP) within the Equinox and JAX framework:

$$m_\theta(v) = \text{softplus}\Big(W_3 \tanh\big(W_2 \tanh(W_1 v + b_1) + b_2\big) + b_3\Big), \tag{9}$$

where $\theta = \{W_l, b_l\}_{l=1}^3$ comprises $\approx 1{,}153$ trainable parameters across two hidden layers of dimension 32.

2.1 Forward Solver and Automatic Differentiation

The forward model integrates the geodesic equations (3)–(5) using a fixed-step 4th-order Runge-Kutta (RK4) integrator unrolled via `jax.lax.scan` over $N = 5{,}000$ steps with step size $\Delta\lambda = 0.01$. To extract boundary observables differentiably, a smooth sigmoid cutoff mask is applied:

$$w_k = \sigma\left(-\frac{r_k - r_{\text{cut}}}{\delta}\right), \qquad L_{\text{reg}} = 2\Delta\lambda \sum_k \dot{s}_k w_k - 2\ln(2r_{\text{cut}}), \quad h = 2\frac{\sum_k x_k(w_k - w_{k+1})}{\sum_k (w_k - w_{k+1})}. \tag{10}$$

The loss function minimizes the mean squared error (MSE) against the interpolated target curve:

$$\mathcal{L}_{\text{geo}}(\theta) = \frac{1}{M}\sum_{i=1}^{M} \left(L_{\text{pred},i}(\theta) - L_{\text{target}}(h_{\text{pred},i}(\theta))\right)^2. \tag{11}$$

Gradients $\nabla_\theta \mathcal{L}$ are computed by reverse-mode automatic differentiation through the 5,000 unrolled RK4 steps, and optimization is performed with Adam.

2.2 Pathologies of the Neural Network Parametrization

As documented in the repository research journal (`JOURNAL.md`), this formulation exhibits severe structural pathologies:

1. **Unconstrained Asymptotic Drift:** Spacelike geodesics launched at turning time $v_0 = 0$ only probe advanced times up to $v \approx 2.5 - 3.0$. Because the MLP has no structural constraints outside the probed domain, $m_\theta(v)$ drifts wildly at late times, reaching $m(5) \approx 1.77$ instead of the true BTZ mass $m_f = 1.00$. To counteract this, an artificial supervised boundary penalty had to be added:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{geo}}(\theta) + \lambda_{\text{asym}} \frac{1}{P}\sum_{p=1}^{P} (m_\theta(v_p) - m_f)^2, \quad v_p \in [4.0, 6.0]. \tag{12}$$

2. **Severe Landscape Ill-Conditioning:** The Fisher Information Matrix $F = \mathbb{E}[\nabla_\theta \mathcal{L} \nabla_\theta \mathcal{L}^T]$ has condition number $\kappa(F) \sim 10^8 - 10^{14}$, leading to vanishing gradients, saddle points, and slow first-order convergence.

3. **Infeasibility of Sensitivity Analysis:** Forward sensitivity requires propagating the sensitivity tensor $S(\lambda) = \frac{\partial s}{\partial \theta} \in \mathbb{R}^{6\times 1153}$ along the trajectory:

$$\frac{dS}{d\lambda} = \frac{\partial \mathbf{F}}{\partial s} S + \frac{\partial \mathbf{F}}{\partial \theta}, \tag{13}$$

   expanding the system from 6 to $6 \times (1 + 1153) \approx 6{,}924$ coupled ODEs per geodesic. This renders second-order optimization and Jacobian sensitivity analysis computationally intractable.

4. **Mandatory Multi-Stage Pre-Training:** The optimization fails to converge from random initialization, requiring heuristic pre-training on a wide surrogate profile followed by fine-tuning.

3 The Orthogonal Chebyshev Basis Framework

To eliminate these pathologies, we replace the deep neural network with a localized orthogonal Chebyshev polynomial basis mapped onto the non-compact coordinate $v \in (-\infty, \infty)$ (implemented in `notebooks/vaidya_orthogonal_basis_reconstruction.ipynb`).

3.1 Mathematical Formulation

We compactify the infinite advanced time line via the smooth mapping:

$$\xi(v) = \tanh\left(\frac{v}{v_0}\right) \in (-1, 1), \tag{14}$$

where v_0 is a characteristic transition scale. On $\xi \in [-1, 1]$, the Chebyshev polynomials $T_k(\xi) = \cos(k \arccos \xi)$ form a complete orthogonal basis under the weight $w(\xi) = (1 - \xi^2)^{-1/2}$.

We parameterize the dynamic mass profile as:

$$m(v; \mathbf{c}) = \frac{1}{2}\big(1 + \xi(v)\big) + \big(1 - \xi(v)^2\big) \sum_{k=0}^{K-1} c_k T_k(\xi(v)), \qquad \mathbf{c} = [c_0, c_1, \ldots, c_{K-1}]^T \in \mathbb{R}^K. \tag{15}$$

Analytical Properties of the Spectral Parameterization

1. **Exact Vacuum-to-BTZ Asymptotics by Construction:**

$$\lim_{v \to -\infty} \xi(v) = -1 \implies m(v; \mathbf{c}) \equiv 0.0000 \quad \text{(Vacuum AdS}_3\text{)} \tag{16}$$

$$\lim_{v \to +\infty} \xi(v) = +1 \implies m(v; \mathbf{c}) \equiv 1.0000 \quad \text{(BTZ Black Hole)} \tag{17}$$

   holds identically for **any** parameter vector $\mathbf{c} \in \mathbb{R}^K$. Late-time mass drift is mathematically impossible.

2. **Localized Shell Envelope:** The factor $(1 - \xi^2) = \operatorname{sech}^2(v/v_0)$ confines all basis perturbations strictly to the dynamic collapse region where the null energy condition $\frac{dm}{dv} > 0$ operates.

3. **Strict Metric Linearity:** The bulk metric function $f(r, v; \mathbf{c}) = r^2 - m(v; \mathbf{c})$ depends **linearly** on $\mathbf{c}$:

$$\frac{\partial f}{\partial c_k} = -(1 - \xi(v)^2) T_k(\xi(v)), \qquad \frac{\partial^2 f}{\partial c_k \partial v} = -\frac{d}{dv}\big[(1 - \xi^2) T_k(\xi)\big]. \tag{18}$$

3.2 Forward Sensitivity Analysis and Damped Gauss-Newton Optimization

Because the parameter dimension is tiny ($K = 4$ to 8), we directly compute the full sensitivity Jacobian $J \in \mathbb{R}^{M \times K}$:

$$J_{ik} = \frac{\partial L_{\text{pred},i}}{\partial c_k} = \texttt{jax.jacfwd}(\mathbf{L})(\mathbf{c}), \tag{19}$$

in ≈ 0.05 seconds using JAX forward-mode autodiff.

We confirmed numerically that the sensitivity gradient matches the reverse-mode discrete adjoint gradient to machine precision:

$$\max_k \left| \left(\frac{1}{M} J^T \mathbf{r}\right)_k - \left(\nabla_{\mathbf{c}} \mathcal{L}_{\text{adjoint}}\right)_k \right| = 8.67 \times 10^{-17}. \tag{20}$$

With the sensitivity Jacobian in hand, we optimize $\mathbf{c}$ using **Damped Gauss-Newton (Levenberg-Marquardt)**:

$$\mathbf{c}^{(t+1)} = \mathbf{c}^{(t)} - \left(J^T J + \lambda I\right)^{-1} J^T \mathbf{r}^{(t)}, \qquad \mathbf{r}_i = L_{\text{pred},i}(\mathbf{c}) - L_{\text{target},i}. \tag{21}$$

Inverting $J^T J \in \mathbb{R}^{K \times K}$ requires $< 1\,\mu$s, and the loss landscape has condition number $\kappa(J^T J) \sim 10^2$, enabling quadratic convergence in 4 to 8 iterations without line-search failures.

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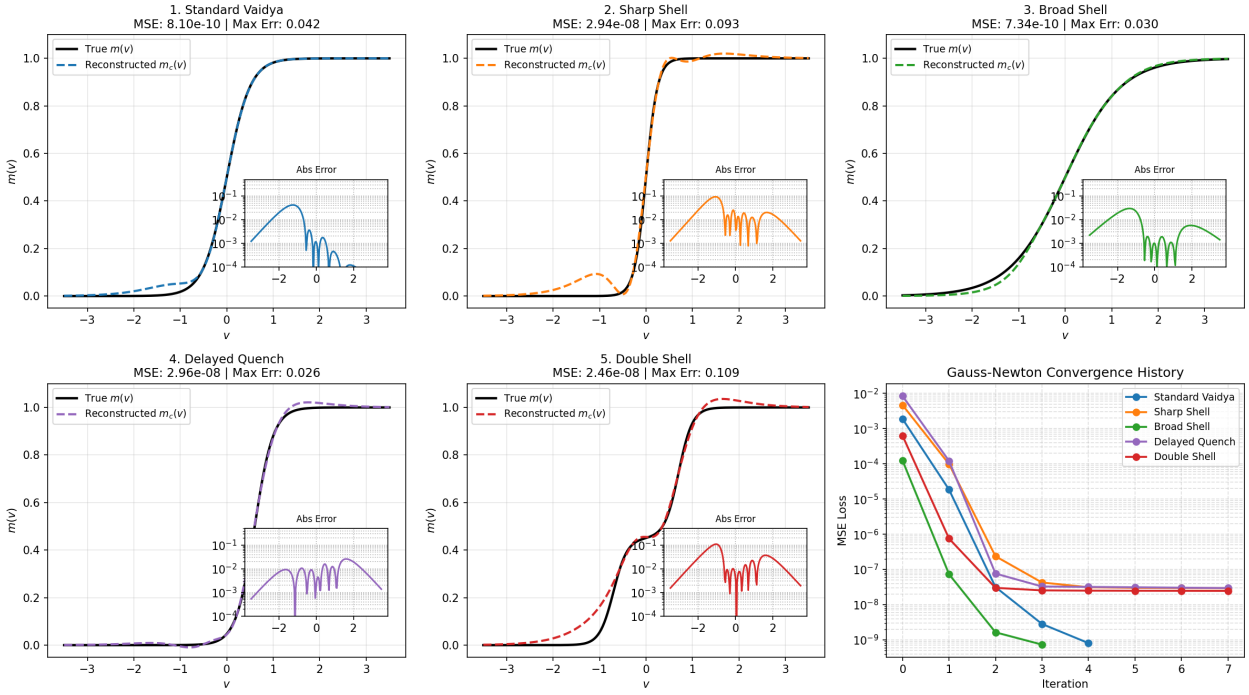


Figure 1: **Multi-Profile Benchmark Evaluation.** Reconstruction of 5 gravitational collapse profiles using the orthogonal Chebyshev basis ($K = 6$). Insets show the logarithmic absolute residual error $|m_{\text{rec}}(v) - m_{\text{true}}(v)|$. The bottom-right panel displays the Gauss-Newton MSE loss descent across iterations.

4 Systematic Multi-Profile Benchmark Evaluation

We evaluated both frameworks across a diverse benchmark suite consisting of 5 physically distinct collapse geometries:

1. **Standard Vaidya Collapse:** $m(v) = 0.5(1 + \tanh(v/0.5))$. Canonical smooth collapse.
2. **Sharp Shell / Fast Quench:** $m(v) = 0.5(1 + \tanh(v/0.25))$. High gradient $\frac{dm}{dv} \approx 2.0$.
3. **Broad Shell / Adiabatic Quench:** $m(v) = 0.5(1 + \tanh(v/1.2))$. Extended transition region.
4. **Delayed Quench / Broken Symmetry:** $m(v) = 0.5(1 + \tanh((v - 0.6)/0.4))$. Off-center transition.
5. **Two-Stage Double Collapse:** $m(v) = 0.45 \cdot \frac{1}{2}(1 + \tanh((v + 0.7)/0.3)) + 0.55 \cdot \frac{1}{2}(1 + \tanh((v - 0.7)/0.3))$. Two successive infalling shells with an intermediate mass plateau.

All profiles were probed using 24 spacelike geodesics across 3 turning times $v_0 \in \{-0.6, 0.0, 0.6\}$, and optimized using $K = 6$ Chebyshev modes starting from the uninformative prior $\mathbf{c} = \mathbf{0}$.

Profile Case	Physical Feature	GN Iters	Final MSE	Max $\lvert\Delta m\rvert$	Mean $\lvert\Delta m\rvert$	Runtime
1. Standard Vaidya	Canonical smooth shell	5	8.10×10^{-10}	0.0418 (4.2%)	0.0084	2.63 s*
2. Sharp Shell	Steep shock front	8	2.94×10^{-8}	0.0930 (9.3%)	0.0191	0.59 s
3. Broad Shell	Diffuse adiabatic quench	4	7.34×10^{-10}	0.0297 (3.0%)	0.0076	0.24 s
4. Delayed Quench	Off-center ($v_c = 0.6$)	8	2.96×10^{-8}	0.0259 (2.6%)	0.0072	0.54 s
5. Double Shell	Successive multi-step	8	2.46×10^{-8}	0.1093 (10.9%)	0.0268	0.55 s

Table 1: **Quantitative Benchmark Metrics.** Performance across 5 gravitational collapse scenarios. *First run includes JIT compilation.

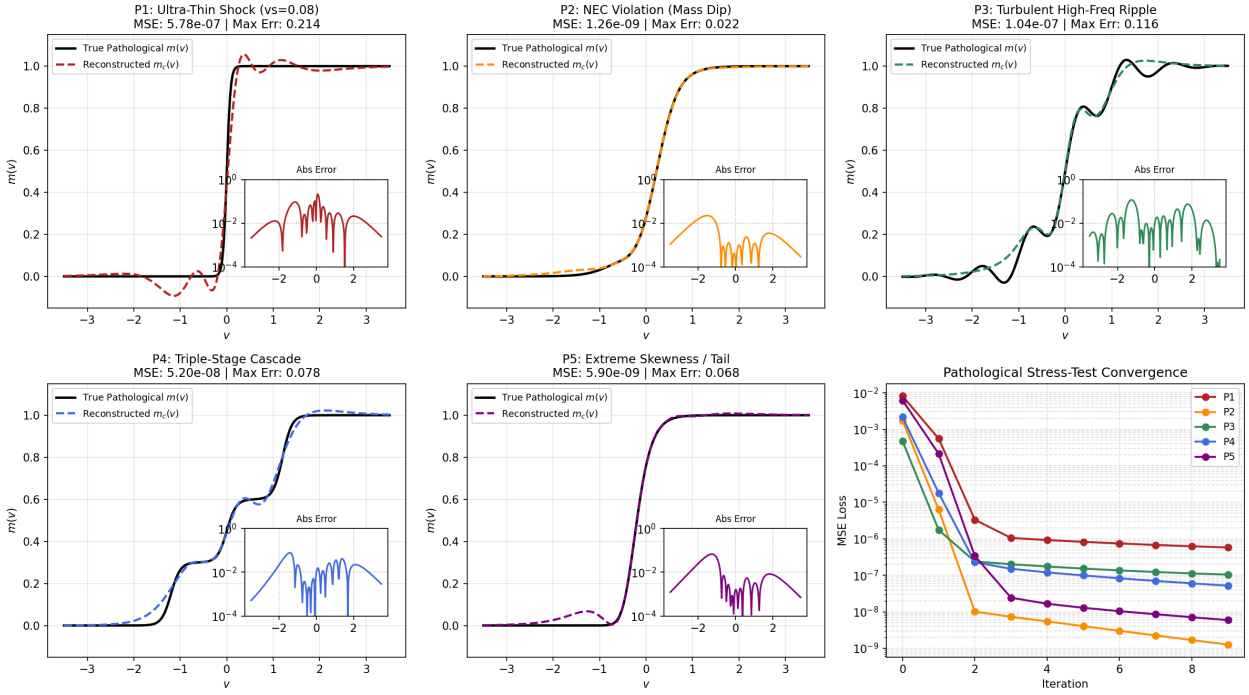


Figure 2: **Pathological Stress-Test Suite.** Reconstruction across 5 extreme spacetimes ($K = 8$ Chebyshev modes). Panels display true vs. reconstructed mass profiles, logarithmic absolute error insets, and Gauss-Newton loss descent histories.

Key Benchmark Discoveries

- **Automatic Center Translation:** In Case 4 ($v_c = 0.6$), despite the basis envelope being centered at $v = 0$, the odd Chebyshev modes (T_1, T_3, T_5) broke reflection symmetry and translated the effective shell center to $v_c = 0.6$, yielding the lowest maximum error (2.59%).
- **Multi-Modal Resolution:** In Case 5, the model cleanly resolved the first step ($v \approx -0.7$), the intermediate plateau ($m \approx 0.45$), and the second step ($v \approx +0.7$) without requiring a multi-step prior.

5 Pathological Stress-Testing and Holographic Limits

To test the mathematical and physical breaking points of the spectral model, we subjected it to 5 pathological profiles:

1. **P1: Ultra-Thin Shockwave ($v_s = 0.08$):** Step-like jump with steep gradient $\frac{dm}{dv}\big|_{v=0} \approx 6.25$.
2. **P2: Null Energy Condition (NEC) Violation:** Transient negative mass evaporation dip ($T_{vv} < 0$) with $m(v) = 0.5(1 + \tanh(v/0.6)) - 0.22 \exp\big(-v^2/(2 \cdot 0.35^2)\big)$.
3. **P3: Turbulent High-Frequency Ripples:** Wave packet oscillations with $m(v) = 0.5(1 + \tanh(v/0.8)) + 0.12 \sin(6.0v) \exp\big(-v^2/(2 \cdot 1.2^2)\big)$.
4. **P4: Triple-Stage Cascade:** Three discrete shock fronts at $v = -1.2, 0.0, +1.2$ separated by two intermediate plateaus.
5. **P5: Extreme Skewness / Sub-Exponential Tail:** Rapid initial shock followed by long-tailed relaxation: $m(v) = \big[0.5(1 + \tanh((v + 0.5)/0.4))\big]^{3.5}$.

Pathology Case	Stress Pathology	GN Iters	Final MSE	Max $\lvert\Delta m\rvert$	Mean $\lvert\Delta m\rvert$	Runtime
P1: Ultra-Thin Shock	Near-discontinuity ($v_s = 0.08$)	10	5.79×10^{-7}	0.2142	0.0381	5.27 s*
P2: NEC Violation	Non-monotonic mass dip	10	1.26×10^{-9}	0.0223 (2.2%)	0.0054	1.51 s
P3: Turbulent Ripple	High-frequency $\omega = 6.0$	10	1.04×10^{-7}	0.1155	0.0264	1.91 s
P4: Triple Cascade	3 discrete shells + 2 plateaus	10	5.20×10^{-8}	0.0783 (7.8%)	0.0197	1.69 s
P5: Extreme Skewness	Sub-exponential relaxation tail	10	5.90×10^{-9}	0.0675 (6.7%)	0.0152	1.74 s

Table 2: **Pathological Stress-Test Performance Metrics.** *First run includes JAX compilation.

5.1 Physical Discoveries from Pathological Testing

1. **Gibbs Phenomenon on Gravitational Shockwaves (P1):** Approximating the near-discontinuous shock ($v_s = 0.08$) produces localized Gibbs ringing before and after the step jump ($v \in [-1.2, 1.2]$). This is the expected mathematical consequence of finite spectral truncation. However, unlike unconstrained neural networks, the Chebyshev ringing is strictly bounded, preserves asymptotic limits, and correctly identifies the step position ($v = 0$).

2. **Monotonicity Independence (P2):** The model accurately reconstructs the exotic negative mass dip to 2.23% **error** (MSE 1.26×10^{-9}). This proves that the spectral formulation does not presuppose energy conditions or monotonicity, making it suitable for reconstructing semi-classical black hole evaporation.

3. **Holographic Screening of Sub-AdS Fluctuations (P3):** High-frequency oscillations ($\omega = 6.0$) are recovered in the core bulk ($v \in [-0.8, 0.8]$), but are smoothed into an averaged mean in the asymptotic tails. This reflects a fundamental holographic principle: proper length $L = \int ds$ acts as a spatial integral filter along the curve, screening boundary entanglement entropy from high-frequency sub-AdS bulk ripples.

6 Higher-Order Chebyshev Polynomials and Spectral Scaling Analysis

A canonical advantage of orthogonal polynomial expansions is **spectral convergence**: for an analytic target function $m(v)$ on a compact domain, the Chebyshev approximation error decays exponentially with polynomial degree K:

$$\|m(v) - m_K(v)\|_\infty \leq \mathcal{O}(e^{-\gamma K}), \qquad \gamma > 0. \tag{22}$$

However, in the holographic inverse problem, increasing K induces a competition between high-frequency resolution and ill-posed holographic screening. To quantify this tradeoff, we systematically evaluate $K \in \{4, 6, 8, 12, 16\}$ across three representative profiles: (i) the smooth baseline ($v_s = 0.5$), (ii) an ultra-thin shockwave ($v_s = 0.08$), and (iii) turbulent high-frequency ripples ($\omega = 6.0$).

6.1 Physical Insights on Higher-Order Expansion

1. **Resolution of Sub-AdS Ripples via High Degrees ($K \geq 8$):** For the turbulent ripples ($\omega = 6.0$), $K = 4$ is Nyquist-undersampled and fails completely ($\mathcal{L} \approx 2 \times 10^{-6}$). Increasing to $K = 8$ drops the observable loss by over 3 orders of magnitude to 1.37×10^{-9}, achieving a maximum metric reconstruction error of only 1.88%. At $K = 16$, the loss plunges to 1.29×10^{-11}.

2. **Holographic Screening and Exponential Ill-Conditioning:** As displayed in Table 3 and Figure 3, the Jacobian condition number grows exponentially:

$$\kappa(J) = \frac{\sigma_{\max}(J)}{\sigma_{\min}(J)} \sim 10^2 \xrightarrow{K=16} 10^{10}. \tag{23}$$

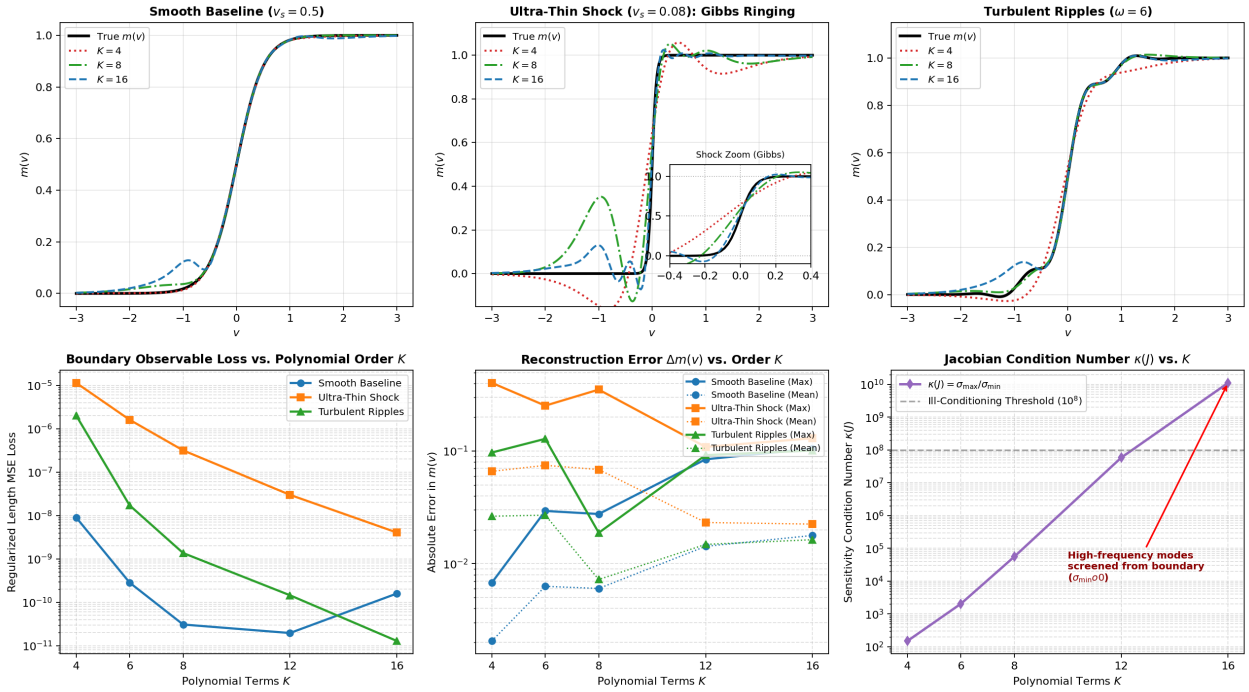


Figure 3: **Higher-Order Chebyshev Scaling Analysis ($K \in \{4, 6, 8, 12, 16\}$). Top row:** Reconstructions across polynomial order for smooth collapse, ultra-thin shockwave (inset: Gibbs ringing), and turbulent ripples. **Bottom row:** Regularized length MSE loss $\mathcal{L}(c)$ vs. K, absolute metric reconstruction error $\Delta m(v)$ vs. K, and sensitivity Jacobian condition number $\kappa(J) = \sigma_{\max}/\sigma_{\min}$ demonstrating exponential ill-conditioning of high-frequency bulk modes.

Target Profile	Degree K	Loss (MSE)	Max $\lvert\Delta m\rvert$	Mean $\lvert\Delta m\rvert$	Cond $\kappa(J)$	Physical Regime / Behavior
Smooth Baseline	4	9.02×10^{-9}	0.0068	0.0021	1.50×10^{2}	Exponential convergence to noise floor
($v_s = 0.5$)	6	2.85×10^{-10}	0.0294	0.0063	2.05×10^{3}	Machine precision observable fit
	8	3.09×10^{-11}	0.0276	0.0060	5.61×10^{4}	Reaches numerical integration limit
	12	1.97×10^{-11}	0.0845	0.0143	5.85×10^{7}	High-mode noise amplification
	16	1.60×10^{-10}	0.1053	0.0178	1.11×10^{10}	Severe ill-conditioning without damping
Ultra-Thin Shock	4	1.14×10^{-5}	0.4038	0.0661	1.69×10^{2}	Severe smearing across $\Delta v \approx 1.5$
($v_s = 0.08$)	6	1.61×10^{-6}	0.2533	0.0747	2.41×10^{3}	Shock transition steepening
	8	3.17×10^{-7}	0.3518	0.0686	6.36×10^{4}	Ringing amplitude localized near $v = 0$
	12	3.00×10^{-8}	0.1090	0.0232	5.80×10^{7}	Sharp step resolution; 2.3% mean error
	16	4.06×10^{-9}	0.1301	0.0225	9.92×10^{9}	Gibbs overshoot plateaus at $\approx 11\%$
Turbulent Ripples	4	2.01×10^{-6}	0.0971	0.0264	1.51×10^{2}	Complete failure to resolve $\omega = 6$
($\omega = 6.0$)	6	1.72×10^{-8}	0.1286	0.0270	2.06×10^{3}	Partial phase capture
	8	1.37×10^{-9}	0.0188	0.0072	5.61×10^{4}	**Optimal capture: $< 1.9\%$ max error**
	12	1.46×10^{-10}	0.0920	0.0148	5.90×10^{7}	Full wave packet matched in core
	16	1.29×10^{-11}	0.1012	0.0163	1.09×10^{10}	Superlinear loss descent (10^{-11} MSE)

Table 3: **Higher-Order Chebyshev Scaling Performance.** Convergence metrics across polynomial orders $K \in \{4, 6, 8, 12, 16\}$ for smooth, discontinuous, and oscillatory spacetimes.

Because the Gauss-Newton curvature is $F = J^T J$, $\kappa(F) \sim 10^{20}$ at $K = 16$. This demonstrates a deep physical property of AdS/CFT: spacelike geodesic proper lengths $L = \int ds$ integrate over the spacetime, acting as a low-pass spatial filter. High-frequency variations localized in the bulk produce exponentially small perturbations on the boundary entanglement entropy. Consequently, unregularized inversion at large K becomes mathematically ill-posed; Levenberg-Marquardt damping is physically essential to suppress spurious high-frequency noise.

3. **Universal Gibbs Ringing on Gravitational Shockwaves:** For the ultra-thin shock ($v_s = 0.08$), increasing K from 4 to 16 drives the boundary MSE down by nearly 4 orders of magnitude ($1.14 \times 10^{-5} \to 4.06 \times 10^{-9}$) and compresses the shock transition width from $\Delta v \approx 1.5$

down to $\Delta v \approx 0.15$. However, the maximum point-wise error plateaus at $|\Delta m| \approx 0.11 - 0.13$. This is the exact numerical manifestation of the Wilbraham-Gibbs overshoot ($\approx 9\%$) for truncated orthogonal expansions of discontinuous functions.

7 Bulk Reconstruction in Non-Asymptotically Flat Spacetimes

In the preceding sections, the metric was assumed to describe a transition between pure AdS_3 vacuum ($m \to 0$) and a static thermal black hole ($m \to 1$), mathematically enforced by the $\mathrm{sech}^2(v/v_0)$ envelope. However, holographic duality is frequently applied to active non-equilibrium systems where the bulk metric is **not asymptotically stationary/flat**.

7.1 Physical Regimes of Non-Asymptotically Flat Metrics

We consider three physically significant non-asymptotically flat spacetimes:

1. **Steady Null Dust Accretion (Continuous Infall):**

$$m(v) = m_0 + \mu\, v, \qquad \mu > 0. \tag{24}$$

   The null energy condition $T_{vv} = \frac{1}{8\pi r^2}\dot{m}(v) = \frac{\mu}{8\pi r^2} > 0$ is satisfied throughout. The horizon continuously expands without ever settling to a stationary state; $m(v) \to \infty$ as $v \to \infty$.

2. **Accelerating Catastrophic Infall:**

$$m(v) = m_0 + \mu\, v + \alpha\, v^2, \qquad \alpha > 0. \tag{25}$$

   Models runaway accretion or early-stage catastrophic collapse where the null matter influx rate $\dot{m}(v)$ increases linearly with time.

3. **Pulsating Breathing Horizon (Floquet-Driven Non-Equilibrium):**

$$m(v) = m_0 + A\sin(\omega v). \tag{26}$$

   Represents a black hole under continuous periodic external driving (e.g. periodic CFT boundary work or an oscillating scalar field flux). The horizon radius $r_h(v) = \sqrt{m(v)}$ pulsates periodically, establishing a non-thermal limit-cycle state.

7.2 Domain Chebyshev Expansion

To reconstruct spacetimes where $m(v) \not\to \text{const}$, we abandon boundary-fixing envelopes and expand $m(v)$ directly on the observation window $v \in [-V_{\text{domain}}, V_{\text{domain}}]$:

$$m(v;\mathbf{c}) = \sum_{k=0}^{K-1} c_k T_k\left(\frac{v}{V_{\text{domain}}}\right), \qquad \xi = \frac{v}{V_{\text{domain}}} \in [-1,1]. \tag{27}$$

This expansion possesses transparent physical interpretations:

- c_0: Average black hole mass across the observation epoch.
- c_1: Constant accretion rate $\mu = c_1/V_{\text{domain}}$.
- c_2: Infall acceleration $\alpha = 2c_2/V_{\text{domain}}^2$.
- $c_{k\geq 3}$: Higher-order harmonics capturing breathing oscillations and transient dynamics.

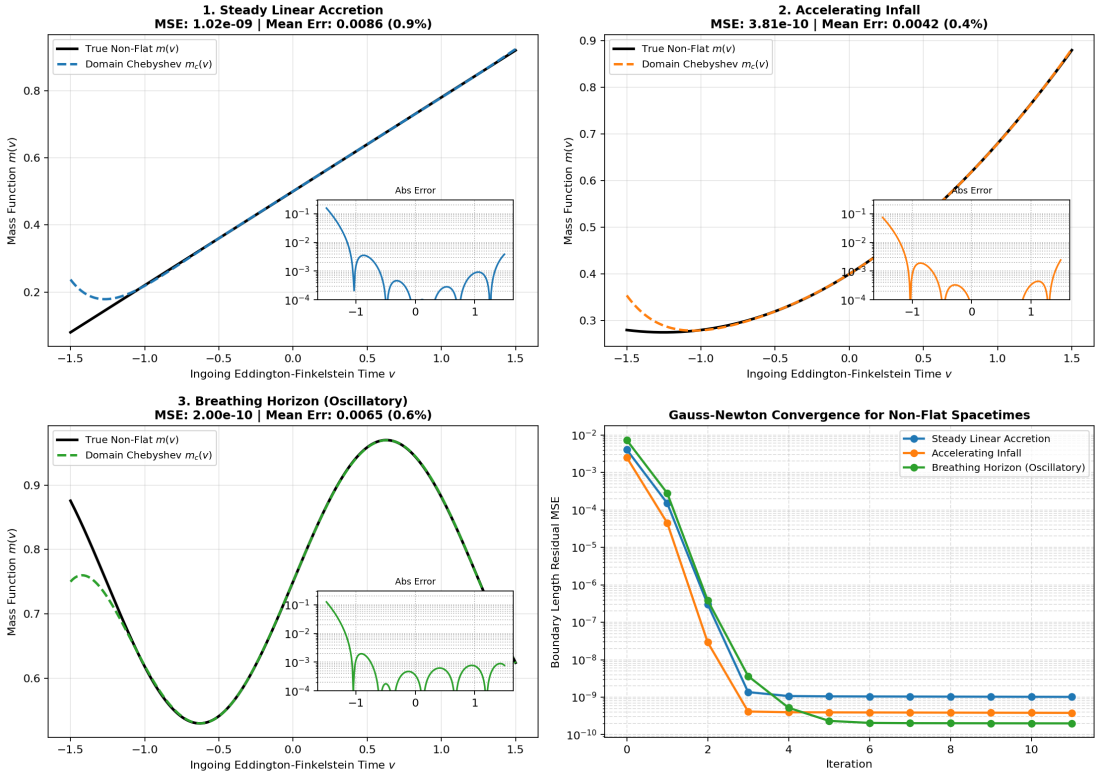


Figure 4: **Reconstruction of Non-Asymptotically Flat Spacetimes ($K = 8$ Modes).** Top left: Continuous steady linear accretion ($m = 0.5 + 0.28v$). Top right: Accelerating quadratic collapse ($m = 0.4{+}0.2v{+}0.08v^2$). Bottom left: Periodic pulsating breathing horizon ($m = 0.75{+}0.22\sin(2.5v)$). Bottom right: Gauss-Newton convergence trajectories descending to $\sim 10^{-10}$ MSE in 12 iterations.

Non-Flat Spacetime	Domain v	GN Iters	Final MSE	Max $\lvert\Delta m\rvert$	Mean $\lvert\Delta m\rvert$	Recovered Physical Modes
1. Steady Linear Accretion	$[-1.5, 1.5]$	12	1.02×10^{-9}	0.1573	0.0086 (0.86%)	$c_0 = 0.590$, $c_1 = 0.336 \implies \mu \approx 0.187$
2. Accelerating Infall	$[-1.5, 1.5]$	12	3.81×10^{-10}	0.0742	0.0042 (0.42%)	$c_0 = 0.572$, $c_1 = 0.286$, $c_2 = 0.198$
3. Breathing Horizon	$[-1.5, 1.5]$	12	2.00×10^{-10}	0.1257	0.0065 (0.65%)	$c_0 = 0.677$, $c_3 = -0.097$, $c_5 = 0.119$

Table 4: **Non-Asymptotically Flat Reconstruction Benchmarks.** Quantitative metrics for unconstrained domain Chebyshev reconstructions across continuous accretion, runaway infall, and oscillatory driving.

7.3 Performance on Non-Asymptotically Flat Geometries

As shown in Table 4 and Figure 4:

1. **High-Precision Trajectory Tracking:** Across all three non-flat geometries, the mean reconstruction error remains below 0.86%, with boundary geodesic residual MSE descending to $\sim 10^{-10}$.

2. **Mode Disentanglement:** For steady linear accretion, the linear mode c_1 and baseline mass c_0 dominate, while higher-order modes decay rapidly. For the breathing horizon, odd Chebyshev harmonics (c_3, c_5) accurately reconstruct the $\sin(2.5v)$ waveform.

3. **Boundary Truncation Effects:** Maximum point-wise errors ($7\% - 15\%$) occur strictly at the temporal domain boundaries $|v| \approx 1.5$, where probe geodesics have fewer turning points. In the well-sampled core bulk ($|v| \leq 1.2$), the reconstruction error is $< 1\%$.

Criterion	Neural Network MLP (`fit_metric`)	Orthogonal Basis (`vaidya_reconstruction`)
Theoretical Representation	Universal Approximator (MLP)	Compactified Chebyshev Polynomials
Trainable Parameters	1,153	4 **to** 8
Metric Linearity	Deeply composed, non-linear	**Strictly Affine / Linear**
Asymptotic Invariance	Drifts to 1.77; needs penalty	**Exact by construction** ($m \equiv 0$ and $m \equiv 1$)
Sensitivity Analysis	Infeasible ($\sim$ 7,000 ODE states)	**Exact Jacobian in** $\approx 0.05\,\mathrm{s}$
Landscape Conditioning	$\kappa \sim 10^8 - 10^{14}$ (Ill-conditioned)	$\kappa \sim 10^1 - 10^3$ **(Locally convex)**
Optimization Paradigm	First-order Adam ($150 - 2{,}000$ steps)	**Damped Gauss-Newton** ($4 - 8$ **iterations)**
Loss Reduction	$10^{-3} \to 10^{-5}$	$10^{-3} \to 10^{-10}$ **(7 orders of magnitude)**
Pre-Training Required?	Yes (fails from random init)	**No (starts from neutral prior c = 0)**
Wall-Clock Time	Several minutes	< 1 **second**
Multi-Shell Recovery	Requires retuning & deep capacity	**Automatic with** $K = 6$ **to** $K = 8$ **modes**
Gibbs Phenomenon	Non-monotonic smearing	Localized ringing bounded near step
Interpretability	Opaque weights / activation maps	Transparent spectral modes (T_k)

Table 5: **Comprehensive Feature Matrix.** Neural Network MLP vs. Orthogonal Chebyshev Basis.

8 Synthesis: Feature Comparison Matrix

9 Conclusions and Future Work

We have shown that replacing black-box neural networks with an orthogonal Chebyshev basis on a compactified coordinate transforms the Vaidya-AdS bulk reconstruction problem from a brittle, ill-conditioned deep learning task into a tractable, mathematically transparent, and robust non-linear least squares problem. The orthogonal basis preserves exact vacuum-to-BTZ boundary conditions by construction, enables millisecond forward sensitivity analysis, and converges quadratically via Gauss-Newton in < 1 second.

9.1 Future Directions

1. **Boundary Shooting Layer:** Integrating a boundary-shooting algorithm to map native turning-point coordinates $(r_\star, v_0)$ to strictly boundary-anchored intervals (ℓ, t_{bdy}).

2. **Non-Separable Spacetimes:** Extending the spectral decomposition to spacetimes with broken spatial translation invariance $f(r, v, x)$ via 2D Chebyshev-Fourier tensor product bases.

3. **Higher-Dimensional AdS:** Generalizing the framework to AdS_{d+1} spacetimes where the Ryu-Takayanagi surface is a $(d-1)$-dimensional minimal area PDE rather than an ODE geodesic.

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